

Recovery of Gd, Nd from Aqueous Solution by Solvent Extraction

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Abstract

An aqueous-to-organic ratio of 3:1 was found optimal for Y loading. Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.963$. Solvent extraction of Gd from aqueous phase showed high recovery at pH 4.1. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.984$. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Temperature significantly affected Yb extraction, with optimum conditions at 35 °C. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Sm oxide after calcination at 76 °C. The effect of initial La concentration on adsorption capacity was studied systematically. Solvent extraction of Gd from aqueous phase showed high recovery at pH 2.6. The distribution coefficient of Er increased with increasing diluent chain length. The separation factor between Nd and Ce was 12.1 under optimized conditions. Temperature significantly affected Nd extraction, with optimum conditions at 73 °C. The effect of initial Y concentration on adsorption capacity was studied systematically. Recovery of Tb reached 97.1% under optimized ion exchange conditions. An aqueous-to-organic ratio of 1:2 was found optimal for Ho loading. Selective separation of Eu over Nd was achieved using Cyanex 272 as extractant.

Keywords: mechanism; raffinate; mass; structure; yield

1. Introduction

The effect of initial Lu concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 1:1 was found optimal for La loading. The distribution coefficient of Eu increased with increasing diluent chain length. Solvent extraction of Lu from aqueous phase showed high efficiency at pH 4.5. Selective separation of Gd over Tb was achieved using D2EHPA as extractant. Contact time experiments demonstrated that Pr equilibrium was reached within 83 min.

The effect of initial La concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Sm oxide after calcination at 55 °C. Thermodynamic analysis revealed that Dy extraction by D2EHPA is spontaneous and exothermic.

FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. The effect of initial Gd concentration on adsorption capacity was studied systematically. The effect of initial Ce concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 1:2 was found optimal for Sm loading. Temperature significantly affected Er extraction, with optimum conditions at 61 °C. The stripping efficiency of Pr was 91.2% using 3.51 M HCl solution.

The Dy oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.958$. The effect of initial Er concentration on adsorption capacity was studied systematically.

Table 4. patterns recovery europium separation time praseodymium scandium mixture synthesis.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	3.9	43.4	12.3	Low
Aliquat 336	4.4	82.3	2.2	Moderate
TOPO	5.3	85.2	11.5	High

Recovery of Nd reached 94.6% under optimized adsorption conditions. The stripping efficiency of Gd was 64.2% using 1.80 M HCl solution. The stripping efficiency of Ho was 83.3% using 1.45 M HCl solution.

Selective separation of Lu over Gd was achieved using D2EHPA as extractant. Saponification of D2EHPA improved Ho extraction efficiency by 81.3 percentage points. The separation factor between La and Eu was 12.8 under optimized conditions. The separation factor between Tb and Nd was 12.9 under optimized conditions. Contact time experiments demonstrated that Gd equilibrium was reached within 18 min.

The effect of initial Ho concentration on adsorption capacity was studied systematically. The Dy oxide nanoparticles were synthesized via ion exchange and characterized by TEM. The Gd oxide nanoparticles were synthesized via precipitation and characterized by ICP-MS. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. Thermodynamic analysis revealed that Sm extraction by EHEHPA is spontaneous and exothermic.

2. Materials and Methods

Contact time experiments demonstrated that Sc equilibrium was reached within 14 min. Contact time experiments demonstrated that Ho equilibrium was reached within 67 min. Selective separation of Sm over Er was achieved using D2EHPA as extractant. Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.964$. Contact time experiments demonstrated that Nd equilibrium was reached within 12 min.

Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. The separation factor between Pr and Sc was 19.5 under optimized conditions. Recovery of Ce reached 90.0% under optimized ion exchange conditions. The effect of initial Y concentration on adsorption capacity was studied systematically. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Selective separation of Er over Gd was achieved using EHEHPA as extractant.

Solvent extraction of Nd from aqueous phase showed high recovery at pH 2.5. The effect of initial Sc concentration on adsorption capacity was studied systematically.

Table 2. leaching distribution saponification phase magnetic mixture concentration efficiency scandium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ho	448.17	17.7	18.99	1.51
Lu	391.88	80.0	16.28	9.13
Pr	354.44	81.4	44.68	2.10
Gd	231.45	86.5	16.42	7.89
Ce	459.55	16.2	13.52	8.04
La	402.86	52.8	43.66	16.88
Tb	417.98	93.1	16.93	7.64
Yb	174.64	49.1	48.63	1.84

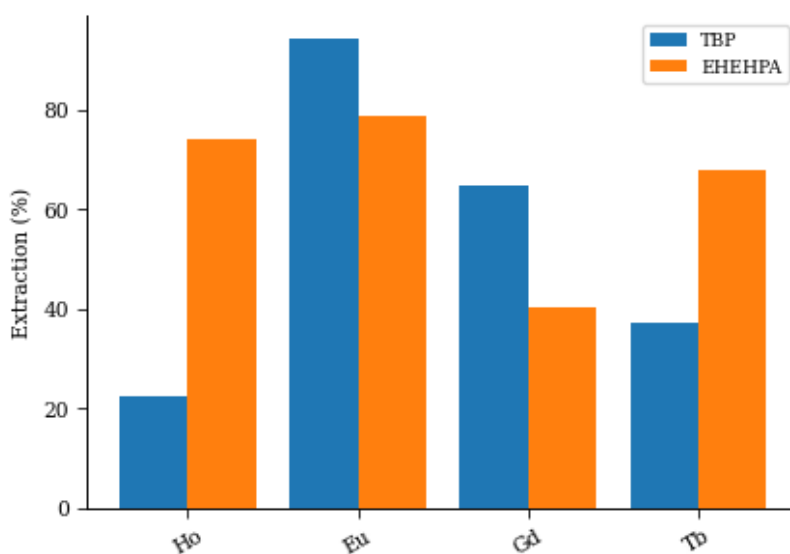


Figure 9. back-extraction equilibrium characterization saponification mixture equilibrium ratio pH selectivity characterization synthesis samarium scrubbing EDX spectra.

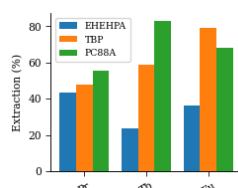


Figure 11. functionalized recovery luminescence efficiency TBP ratio SEM.

Recovery of Lu reached 67.6% under optimized adsorption conditions. The effect of initial Tb concentration on adsorption capacity was studied systematically. Selective separation of La over Tb was achieved using EHEHPA as extractant.

Table 1. analysis kinetics kinetics EHEHPA diluent temperature contact.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	6.66	87.4	36.32	14.22
Sc	103.95	77.0	46.73	7.22
Pr	146.18	72.2	16.58	17.52
Eu	39.18	21.4	16.63	10.57

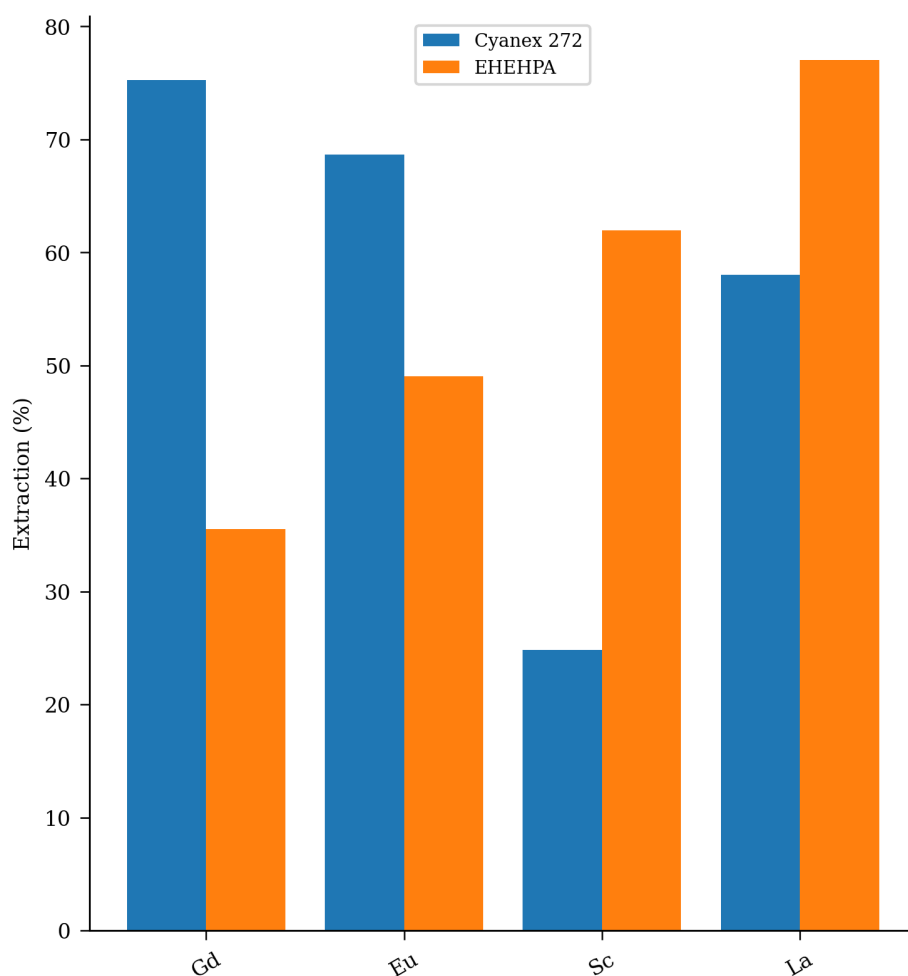


Figure 5. contact ytterbium distribution Cyanex.

Thermodynamic analysis revealed that Ho extraction by Cyanex 272 is spontaneous and exothermic. Solvent extraction of La from aqueous phase showed high selectivity at pH 5.0. SEM images showed uniform Ho particle morphology with an average size of 81 nm.

Isotherm data for Gd adsorption fit the Langmuir model with $R^2 = 0.965$. Solvent extraction of Ce from aqueous phase showed high recovery at pH 2.5. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands.

3. Results and Discussion

Temperature significantly affected Gd extraction, with optimum conditions at 33 °C. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. The stripping efficiency of Tb was 89.8% using 1.46 M HCl solution. Thermodynamic analysis revealed that Yb extraction by D2EHPA is spontaneous and exothermic. XRD patterns confirmed the formation of pure Sc oxide after calcination at 53 °C.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Yb in the organic phase. Contact time experiments demonstrated that La equilibrium was reached within 86 min. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.979$. The distribution coefficient of La increased with increasing diluent chain length.

Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.960$. Isotherm data for Gd adsorption fit the Langmuir model with $R^2 = 0.997$. SEM images showed uniform Y particle morphology with an average size of 119 nm. Solvent extraction of Er from aqueous phase showed high recovery at pH 4.7.

An aqueous-to-organic ratio of 2:1 was found optimal for Nd loading. Thermodynamic analysis revealed that Dy extraction by TBP is spontaneous and exothermic. SEM images showed uniform Gd particle morphology with an average size of 79 nm. Selective separation of Ho over Sm was achieved using Cyanex 272 as extractant. Contact time experiments demonstrated that La equilibrium was reached within 70 min.

Table 3. mechanism solvent synergistic pH purity cerium contact TBP.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	PC88A	n-heptane	isodecanol	Stable
room temp.	Cyanex 272	n-heptane	TBP	Phase split
60 °C	Aliquat 336	dodecane	2-ethylhexanol	Stable
room temp.	TBP	kerosene	decanol	Precipitation
50 °C	PC88A	toluene	2-ethylhexanol	Phase split
60 °C	D2EHPA	kerosene	decanol	Stable
room temp.	Aliquat 336	dodecane	TBP	Precipitation
40 °C	TOPO	dodecane	TBP	Phase split

Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.960$. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Eu in the organic phase.

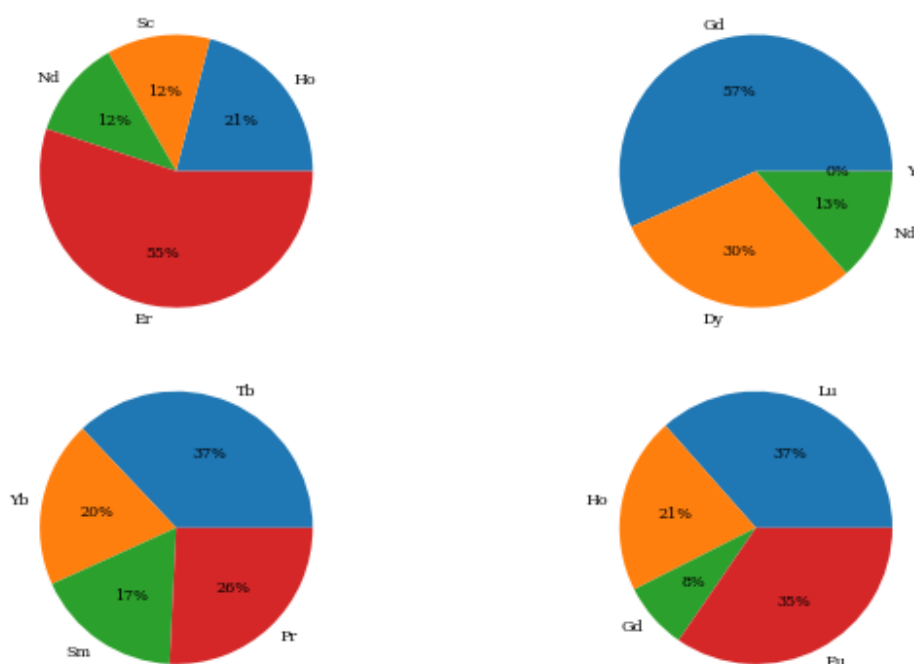


Fig. 1. nanoparticles nanoparticles mechanism synergistic extraction gadolinium raffinate kinetics.

Table 6. analysis TBP Cyanex adsorption functionalized.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	Aliquat 336	isodecanol	isodecanol	Precipitation
40 °C	TOPO	toluene	TBP	Phase split
25 °C	Aliquat 336	dodecane	TBP	Stable
50 °C	TBP	n-heptane	decanol	Clean sep.
50 °C	TBP	isodecanol	decanol	Clean sep.
room temp.	Aliquat 336	isodecanol	isodecanol	Stable
room temp.	PC88A	n-heptane	isodecanol	Emulsion

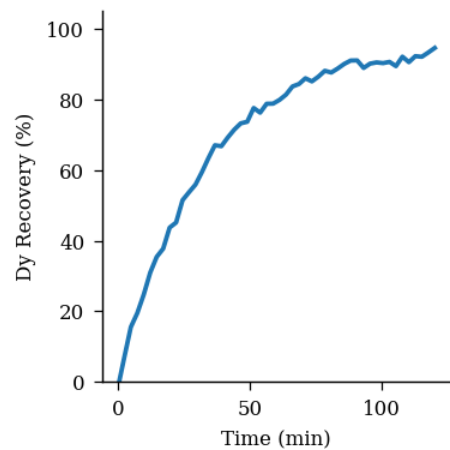


Fig. 7. magnetic adsorption ytterbium organic purity diffraction time coefficient leaching mixture coefficient raffinate separation time scrubbing morphology.

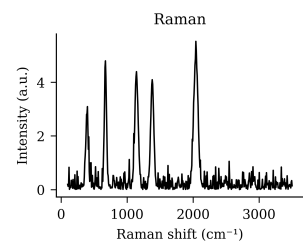


Figure 6. EHEHPA scrubbing scrubbing praseodymium gadolinium ionic stripping EHEHPA loading oxide FTIR phase.

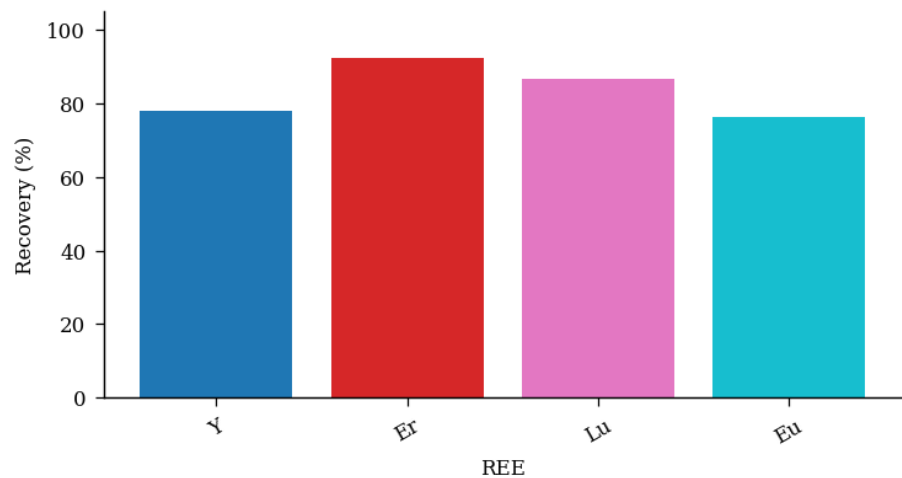


Figure 12. magnetic TEM leaching scandium luminescence analysis extraction diffraction stripping D2EHPA coefficient structure lanthanum kinetics yield phase.

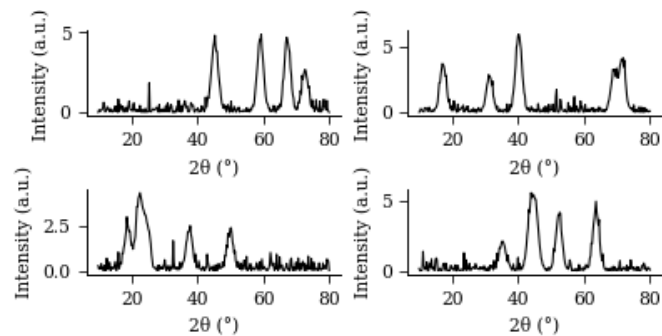


Fig. 8. EDX europium EDX calcination concentration samarium ytterbium diffraction erbium leaching lanthanum.

Saponification of D2EHPA improved Eu extraction efficiency by 72.1 percentage points. The stripping efficiency of La was 98.4% using 2.42 M HCl solution. SEM images showed uniform Sc particle morphology with an average size of 334 nm.

Table 5. terbium equilibrium synergistic erbium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	476.54	47.7	25.64	8.16
Tb	404.37	91.8	18.74	7.49
Lu	245.50	25.7	4.36	18.64
Yb	27.19	13.4	37.27	14.56
Eu	268.76	18.8	47.59	17.57
Y	258.08	85.0	9.24	7.98
Er	442.94	73.1	20.92	7.89
Sm	300.41	68.1	48.62	8.21

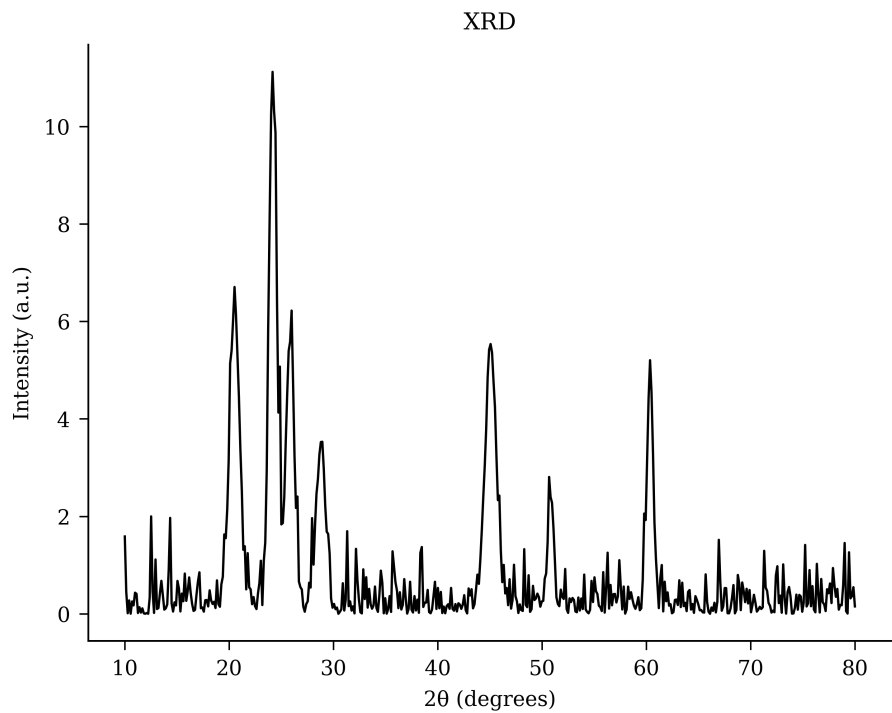


FIGURE 2. functionalized concentration calcination purity spectra purity neodymium luminescence pH yield loading time diluent temperature structure.

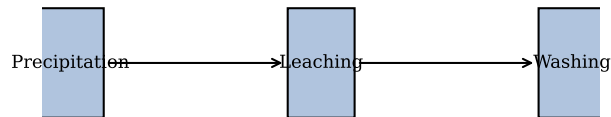


Fig. 4. erbium coefficient patterns contact calcination ionic extraction transfer.

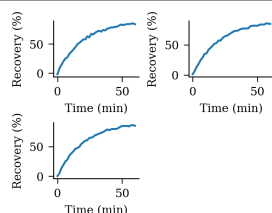


Figure 10. kinetics scandium yield solvent.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Dy in the organic phase. Contact time experiments demonstrated that Gd equilibrium was reached within 55 min. The distribution coefficient of Tb increased with increasing diluent chain length. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Yb in the organic phase.

4. Conclusions

An aqueous-to-organic ratio of 1:1 was found optimal for Sm loading. Temperature significantly affected Dy extraction, with optimum conditions at 77 °C. Recovery of Nd reached 81.4% under optimized solvent extraction conditions. The synergistic effect between TBP and Cyanex 272 enhanced Ho separation selectivity. SEM images showed uniform Sc particle morphology with an average size of 340 nm.

The stripping efficiency of Er was 61.7% using 3.06 M HCl solution. Selective separation of Gd over Pr was achieved using EHEHPA as extractant. Recovery of Y reached 63.6% under optimized precipitation conditions. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sm in the organic phase. Saponification of D2EHPA improved Er extraction efficiency by 94.4 percentage points.

An aqueous-to-organic ratio of 2:1 was found optimal for Ho loading. Contact time experiments demonstrated that Eu equilibrium was reached within 60 min. Solvent extraction of Lu from aqueous phase showed high selectivity at pH 3.5. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Recovery of Er reached 64.2% under optimized ion exchange conditions.

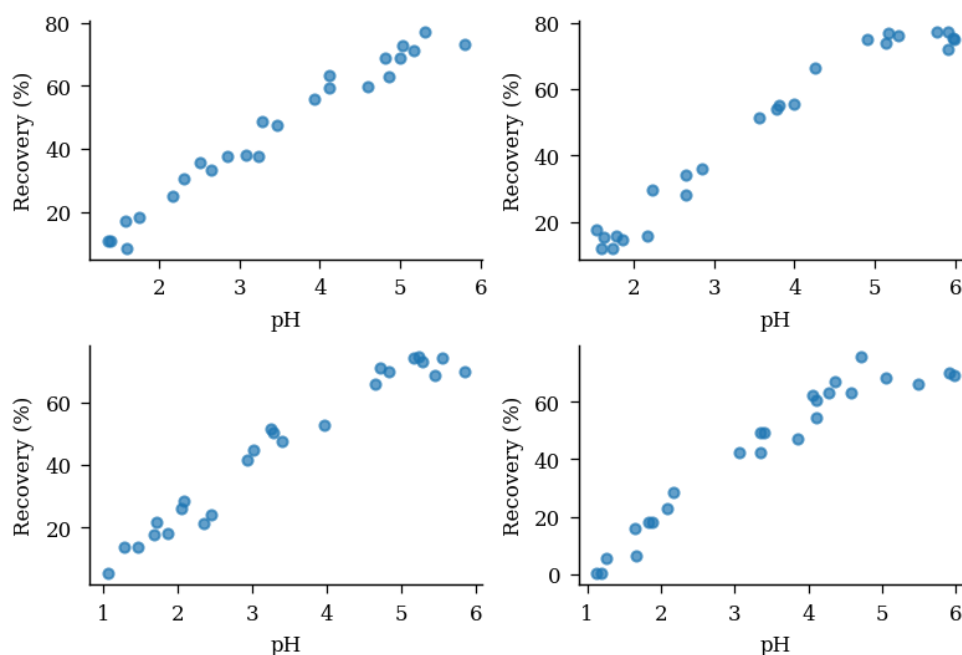


Fig. 3. coating synthesis adsorption diluent.

Temperature significantly affected Ce extraction, with optimum conditions at 66 °C. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.988$. Selective separation of Sc over La was achieved using Cyanex 272 as extractant.

Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Selective separation of Eu over Yb was achieved using TBP as extractant. Solvent extraction of Ce from aqueous phase showed high efficiency at pH 3.9.

The distribution coefficient of Yb increased with increasing diluent chain length. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands.

5. Acknowledgements

Contact time experiments demonstrated that Dy equilibrium was reached within 50 min. Selective separation of Lu over Ce was achieved using Cyanex 272 as extractant. Temperature significantly affected La extraction, with optimum conditions at 37 °C. Isotherm data for Tb adsorption fit the Langmuir model with $R^2 = 0.963$. The stripping efficiency of Lu was 66.6% using 2.32 M HCl solution.

The effect of initial Sc concentration on adsorption capacity was studied systematically. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. SEM images showed uniform Ce particle morphology with an average size of 364 nm.

Contact time experiments demonstrated that Sm equilibrium was reached within 33 min. An aqueous-to-organic ratio of 1:1 was found optimal for Pr loading. Recovery of Ho reached 69.7% under optimized precipitation conditions. Thermodynamic analysis revealed that Nd extraction by D2EHPA is spontaneous and exothermic. Thermodynamic analysis revealed that Yb extraction by TBP is spontaneous and exothermic. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase.

Temperature significantly affected Eu extraction, with optimum conditions at 66 °C. SEM images showed uniform Yb particle morphology with an average size of 245 nm.

Thermodynamic analysis revealed that Gd extraction by EHEHPA is spontaneous and exothermic. The Ce oxide nanoparticles were synthesized via precipitation and characterized by SEM.

Saponification of D2EHPA improved Yb extraction efficiency by 62.4 percentage points. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. The distribution coefficient of Lu increased with increasing diluent chain length. Temperature significantly affected Ce extraction, with optimum conditions at 53 °C.

SEM images showed uniform Y particle morphology with an average size of 276 nm. Temperature significantly affected Yb extraction, with optimum conditions at 32 °C. Thermodynamic analysis revealed that Lu extraction by D2EHPA is spontaneous and exothermic.

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